The Volume Inflection Model

Stationary four-volume in an anisotropic compactification

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Abstract

We specify a homogeneous solution of five-dimensional Einstein gravity with three equal external lengths and a compact circle. Its constant positive density and negative inverse-fifth-power density select a radius at which the spatial four-volume falls through a stationary inflection. We prove a local existence criterion, determine the energy-condition violation it requires and bound the finite lifetime if the stress law is continued. Five prescribed volume readings force opposite signs of curvature of log-volume as a function of log external length. In the larger class of analytic radius-dependent densities, every finite set of tuned volume readings in the stated window also admits a history with strictly negative slope throughout. One calibrated spectral line identifies the change in the circle-to-external length ratio. A second stable standard would recover the relative volume. We quantify how reference drift changes finite-reading curvature tests. The evolving-torus stress, matter couplings and existence of the event in cosmic history remain physically unestablished.

1. Model and independent inputs

Let $\mathcal{M} = I \times T^3 \times S^1$, where I is a proper-time interval. Each external coordinate z^i has unit period and the angular coordinate y has period 2π. In units $c = \hbar = 1$ the metric is

$$ds^2 = -dt^2 + a^2(t)\delta_{ij}dz^i dz^j + b^2(t)dy^2, \qquad a, b > 0.$$

The foliation is orthogonal to the homogeneous comoving congruence, and t measures its proper time. Each principal spatial direction closes on a fixed-time slice. Causal traversal of a full period also requires sufficient elapsed time (§8). Each constant-t slice is a compact flat four-manifold. Its three equal torus lengths are a, the circle circumference is $2\pi b$, and the proper spatial four-volume is $V = 2\pi a^3 b$. Define $H = \dot{a}/a$, $K = \dot{b}/b$ and $\theta = \dot{V}/V = 3H + K$.

Use $G_{AB} = \kappa T_{AB}$ with $\kappa > 0$. The sources are a constant density $\rho_\Lambda > 0$ with isotropic pressure $-\rho_\Lambda$, and the homogeneous prescription

$$\rho_C = -C/b^5, \qquad p_{x,C} = -\rho_C, \qquad p_{y,C} = 4\rho_C, \qquad C > 0.$$

The energy per unit external proper three-volume of the second source is $2\pi b\rho_C$, so differentiation in the circle length gives $p_{y,C} = -d(b\rho_C)/db$. Conservation gives $\dot{\rho}_C = -5K\rho_C$. In particular $\rho + p_y = 5\rho_C < 0$ violates the null energy condition in the compact direction.

Specify $\kappa, \rho_\Lambda, C > 0$. For the tuned expanding branch the equations will fix

$$b_*^5 = \frac{3C}{8\rho_\Lambda}, \qquad H_*^2 = \frac{5\kappa\rho_\Lambda}{18}, \qquad K_* = -3H_*.$$

The external period $a_* > 0$ is free. A time translation fixes the label t_*. Dimensionless plots use these event scales; their normalization assigns no physical age in years.

Preliminary cosmological model, without external peer review or a fit to cosmological data. Mathematical and computational assistance by ChatGPT (OpenAI).

2. Existence and nature of the event

For spatially flat homogeneous slices the Hamiltonian constraint and the two distinct spatial equations are

$$3H(H+K) = \kappa(\rho_\Lambda + \rho_C), \tag{1}$$

$$\dot{H} + \theta H = \kappa(2\rho_\Lambda/3 - \rho_C), \qquad \dot{K} + \theta K = \kappa(2\rho_\Lambda/3 + 4\rho_C). \tag{2}$$

These follow from $R_{00} = -\dot{\theta} - 3H^2 - K^2$ and $R_{ii} = \dot{H}_i + \theta H_i$ in an orthonormal frame. The residual F of (1), left minus right, obeys $\dot{F} = -2\theta F$ by (2) and conservation.

Theorem 1. Stationary inflection on the expanding branch

For the positive constants of §1 and any $a_* > 0$, the event data in §1 determine a unique local analytic solution on which $\dot{V}_* = \ddot{V}_* = 0$ and $V'''(t_*)/V_* = -144H_*^3$. Within this source law an externally expanding interval of constant volume is impossible.

Proof. Summing (2) gives

$$\ddot{V}/V = \dot{\theta} + \theta^2 = \kappa(8\rho_\Lambda/3 - C/b^5).$$

At a stationary event $K = -3H$. Zero acceleration therefore fixes b_*, and (1) then fixes $H_* > 0$. These data obey (1). Local existence and uniqueness follow from the analytic evolution equations at $a, b > 0$. As the circle contracts through b_*, $\ddot{V}$ changes from positive to negative. Its derivative at the event gives $V'''/V_* = 5\kappa K_* C/b_*^5 = -144H_*^3$. Thus $\dot{V}$ attains a strict local maximum of zero and is negative on both sides. On an interval with $H > 0$ and constant V, the same acceleration identity would fix $b = b_*$ throughout, contradicting $K = -3H < 0$. ∎

For any regular increasing change of time on the *same slices*, $(\dot{V}, \ddot{V}) = (0, 0)$ is invariant by the chain rule. Its leading nonzero derivative is rescaled. A different foliation or a different volume observable requires its own specification (§6).

Local coefficients and units

Set $x = \log(a/a_*)$, $R = \log(V/V_*)$, $u = \kappa\rho_\Lambda/(3H^2)$ and $v = \kappa\rho_C/(3H^2)$. From (1), $K/H = u + v - 1$. With prime denoting d/dx, (2) gives

$$u' = u(4 - 2u + 8v), \quad v' = v(9 - 7u + 3v), \quad R' = 2 + u + v. \tag{3}$$

At $(u, v) = (6/5, -16/5)$, differentiating gives $(R', R'', R''', R^{(4)}) = (0, 0, -144, 6480)$. In x and in dimensionless comoving time $\tau = H_*(t - t_*)$, respectively,

$$R = -24x^3 + 270x^4 + O(x^5), \qquad V/V_* - 1 = -24\tau^3 - 162\tau^4 + O(\tau^5). \tag{4}$$

Indeed $\dot{H}_* = 12H_*^2$ gives $x = \tau + 6\tau^2 + O(\tau^3)$. A rational enclosure of (3) on $r = 0.01$ gives $10^6[R(-r) - R(r)] \in (48.67431, 48.67432)$.

3. Dependence on initial data

Proposition 1. A fold in the homogeneous initial data

Keep C, ρ_Λ, κ fixed. At a common external scale a_* choose $u_0 = 6/5$, $v_0 = -16/5 + \lambda$, with $|\lambda|$ small. The perturbed data remain admissible since $H_0^2 = \kappa\rho_\Lambda/(3u_0)$, $b_0^5 = -\kappa C/(3H_0^2 v_0)$ and (1) fixes K_0. Let $D(x,\lambda) = R'(x,\lambda)$. Equation (3) gives

$$D(0,\lambda) = \lambda, \quad D_x(0,\lambda) = -9\lambda + 3\lambda^2, \quad D_{xx}(0,0) = -144.$$

Since $D_\lambda(0,0) = 1$, the implicit function theorem gives an analytic curve of zeros $\lambda = g(x)$ with $g(0) = g'(0) = 0$ and $g''(0) = 144$. Hence $g(x) = 72x^2 + O(x^3)$ and it is strictly convex near zero. Positive sufficiently small λ gives two stationary epochs

$$x_\pm = \pm\sqrt{\lambda/72} + O(\lambda).$$

Negative sufficiently small λ gives no nearby stationary epoch. The earlier crossing for $\lambda > 0$ is a volume minimum and the later one a maximum. On the positive branch, integrating $D = \lambda - 72x^2 + O(|x|^3 + |\lambda x| + \lambda^2)$ between the roots yields $R(x_+) - R(x_-) = 4\lambda^{3/2}/(3\sqrt{72}) + O(\lambda^2)$. This fold concerns the specified transverse family of homogeneous data.

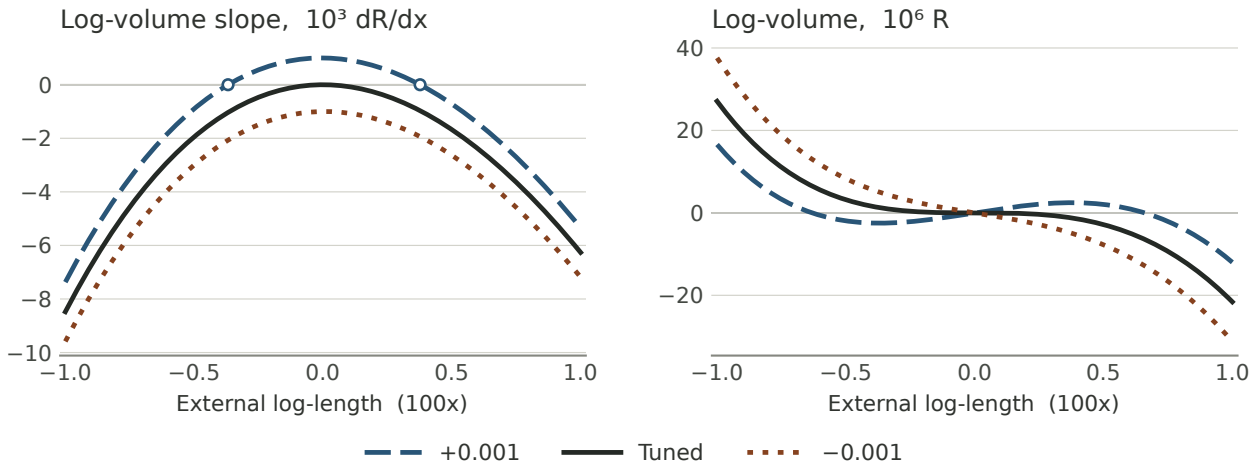


Figure 1. Solutions of (3) on the same interval of external log-length. Line patterns identify the three histories. Dots mark the two turning points for $\lambda = 0.001$. Exact sign bounds place them in $(-0.00368527, -0.00368525)$ and $(0.00376859, 0.00376861)$. For $\lambda = -0.001$, D is strictly negative on the entire displayed interval.

Verification. We compute degree-24 Taylor polynomials with rational coefficients, bound their residual and use a Jacobian row-sum bound and Grönwall's inequality to enclose the solutions. Interval evaluation gives $D_{xx} < -1$ on 40 subintervals for each plotted history. Inhomogeneous perturbations are outside this calculation.

4. Constitutive conditions for stationarity

Retain flat spatial sections, ρ_Λ and external vacuum pressure. Replace the radius-dependent density by a smooth function $f(b)$. Write $\rho = \rho_\Lambda + f$. Conservation for every homogeneous trajectory holds for $p_x = -\rho$ and $p_y = -\rho - bf'(b)$. Here f' denotes a radius derivative. The constraint is still (1), with ρ_C replaced by f. Trace reversal gives

$$\dot{H} + \theta H = \frac{\kappa}{3}(2\rho + bf'), \quad \dot{K} + \theta K = \frac{\kappa}{3}(2\rho - 2bf'), \quad \dot{\theta} + \theta^2 = \frac{\kappa}{3}G(b), \qquad G(b) = 8\rho + bf'.$$

Theorem 2. Local criterion for a general radius-dependent density

Fix $b_* > 0$. An externally expanding event with $R' = R'' = 0$ exists at this radius if and only if $G(b_*) = 0$ and $\rho(b_*) < 0$. Necessity follows from $K_* = -3H_*$, the constraint $H_*^2 = -\kappa\rho(b_*)/6$, and $\dot{\theta}_* = 0$. Conversely these relations specify admissible data and the evolution equations have a unique local solution. At the event $\ddot{\theta}_* = -\kappa H_* b_* G'(b_*)$. With $x = \log(a/a_*)$ this yields

$$R'''(0) = -\frac{\kappa b_* G'(b_*)}{H_*^2}. \tag{5}$$

It is a cubic inflection precisely when this derivative is nonzero. At every event covered by the criterion,

$$\rho_* = -6H_*^2/\kappa < 0, \qquad (\rho + p_y)_* = 8\rho_* = -48H_*^2/\kappa.$$

Indeed $b\rho' = -8\rho$ there. This energy cost holds for every density in the stated class. More generally, for four flat homogeneous directions with rates H_i, $2\kappa\rho = (\sum_i H_i)^2 - \sum_i H_i^2$. A zero total rate with any nonzero individual rate already requires $\rho < 0$. The null-energy conclusion above also uses the pressure law and zero volume acceleration. For a smooth parameter family of stresses close to the inverse-fifth-power law in C^2 near b_*, a simple zero of G persists at a nearby radius by the implicit function theorem. The initial data must be retuned to this radius. Its cubic coefficient depends on the stress derivative. A transverse variation can split the double root as in §3; the sign admitting two roots depends on the cubic coefficient.

A fourth-order test within the two-term family

Allow three external rates H_i with mean H, geometric-mean scale a, equal external pressures, and shear $\sigma^2 = \frac{1}{2}\sum_i (H_i - H)^2$. Let $d = B/(Cb_*)$ for the two-term law in section 5 and $\eta = \sigma_*^2/(-\kappa\rho_*)$. At the retuned simple root the constraint gives $6H_*^2 = -\kappa\rho_* - \sigma_*^2$. Differentiation of the mean and trace equations at the event gives

$$24c_4 = \frac{\kappa}{3H_*^2}[9b^2G'' + (9 + 15\dot{H}/H^2)bG']_*.$$

On the admissible branch with $c_3 = -24$, substitution gives $\eta = -2d/(5 - 6d)$, $d \le 0$, and $c_4 = 270 + 324\eta/(1 - \eta)$. Thus $c_4 = 270$ singles out $B = 0$ and zero shear within this family. The example $d = -1/10$, $\eta = 1/28$ has $c_4 = 282$. A smooth density addition of order $(b - b_*)^4$ leaves G, G', G'' unchanged at the event.

Exact compensation on an interval

If θ vanishes throughout an expanding interval, b varies there and G must vanish on an interval of radii. Integrating $b\rho' = -8\rho$ forces $\rho = B_8 b^{-8}$ with a constant $B_8 < 0$. For local sufficiency choose $K = -3H$ and $H = \sqrt{-\kappa B_8/6}\, b^{-4}$. Then $\dot{b} = -3Hb$ and $\dot{H} = 12H^2$ satisfy both spatial equations. Thus an exactly constant four-volume is possible within this broader homogeneous class only for the specified local density law. The inverse-fifth-power stress fails this criterion.

5. Static benchmark and source sensitivity

In flat $\mathbb{R}^{1,3} \times S^1$, a free, massless, periodic real scalar gives a definite normalization. Set $L = 2\pi b$ and subtract the decompactified vacuum contribution. The nonzero winding terms in the proper-time integral for density per proper four-dimensional spatial volume give

$$f_{\rm stat} = -\frac{1}{2(4\pi)^{5/2}} \sum_{j\neq 0} \int_0^\infty s^{-7/2} e^{-j^2 L^2/(4s)} ds = -\frac{3\zeta(5)}{4\pi^2 L^5}.$$

The integral follows by the substitution $j^2L^2/(4s)$. Thus $C = 3\zeta(5)/(128\pi^7)$ for this single-field static prescription. Multiplying by $2\pi b$ gives the familiar inverse-fourth-power energy per external three-volume [6,8]. On a finite static torus additional windings along the external directions contribute. Here f denotes the finite-torus topological density for this periodic scalar. Virtual work gives $p_x = -f - (a/3)\partial_a f$ with $\partial_a f > 0$. When $a = 2\pi b$, permutation symmetry and homogeneity of degree -5 give $p_x = p_y = f/4$. This disagrees even in sign with $p_{x,C} = -\rho_C$ for the negative prescribed component. The moving torus also requires a state choice and curvature corrections [7]. No gravitational coupling, radius, epoch or line detection is fitted by using this coefficient.

Quantitative source sensitivity

Consider the separate exact prescription $f = -C/b^5 + B/b^6$ with fixed $C > 0$. An inverse-fifth-power contribution per external three-volume would give this second power per spatial four-volume [6]. The value and sign of B require microscopic input. At a retuned event put $X = C/b_*^5$, $d = B/(Cb_*)$. Then $G = 0$ gives $8\rho_\Lambda = (3-2d)X$ and $H_*^2 = \kappa X(5-6d)/48$ for zero external shear. For $d < 5/6$, (5) yields

$$R(x) = -8\,\frac{15-12d}{5-6d}\,x^3 + O(x^4). \tag{6}$$

The cubic coefficients for $d = -0.1, 0, +0.1$ are $-162/7$, -24 and $-276/11$, respectively. The derivative of the cubic coefficient at $d = 0$ is $-48/5$. These values measure sensitivity to the assumed source. The certified fourth-order coefficient and the finite-window tests in sections 2--3 apply only at $d = 0$.

What a potential minimum would establish

For $0 < d < 5/6$, put $z = b/b_*$. The Einstein-frame potential of this two-term law has derivative of the same sign as $-(3-2d)z^6/8 + 6z - 7d$. This strictly concave function is negative at zero, positive at one and negative at infinity. Exactly two positive extrema exist, an inner minimum and an outer maximum. For $d \to 0^+$ the inner root has $z \sim 7d/6$ and $B/(Cb) \to 6/7$ there. The terms are comparable at the minimum in this limit. At b_* the potential is $-6M_*^2H_*^2$. Since it increases from the inner minimum to b_*, the minimum is strictly more negative. The spatially flat Einstein constraint with only this scalar excludes rest there. The tuned solution crosses it in finite time (section 8). A microscopic use of this minimum requires control of omitted stress terms.

Algebra for the shear comparison

At the root $bG' = (15-12d)X$, $b^2G'' = (-90+84d)X$, and $48H_*^2 = \kappa X(5-6d)(1-\eta)$. The mean external equation gives $\dot H_*/H_*^2 = 12/(1-\eta)$. Thus (5) divided by six gives $c_3 = -8(15-12d)/[(5-6d)(1-\eta)]$. Fixing $c_3 = -24$ determines η as in §4. To obtain the quartic formula there, differentiate $\dot\theta + \theta^2 = \kappa G/3$ twice at the event and use $\dot K_* = -3\dot H_*$. Changing from t to x gives $24c_4 = \ddot\theta_*/H_*^4 - 6\dot H_*\ddot\theta_*/H_*^5$. Substitution of these quantities gives the stated $270 + 324\eta/(1-\eta)$.

6. Four-dimensional description and local gravity

For fields independent of y and vanishing Kaluza--Klein vector, the four-dimensional Jordan-frame action of this homogeneous sector is

$$S_J = 2\pi \int d^4z \sqrt{-g_J} \left[\frac{b\,\mathcal{R}_J}{2\kappa} - U(b) \right], \quad U(b) = \rho_\Lambda b - C b^{-4}.$$

Here $\sqrt{-g_5} = b\sqrt{-g_J}$ and $\mathcal{R}_5 = \mathcal{R}_J - 2b^{-1}\nabla^\alpha \nabla_\alpha b$; the last contribution integrates to a boundary term. Variation with respect to the metric and scalar gives

$$bG_{\mu\nu} = \nabla_\mu \nabla_\nu b - g_{\mu\nu} \nabla^\alpha \nabla_\alpha b - \kappa U g_{\mu\nu}, \qquad \mathcal{R}_J = 2\kappa U'(b).$$

The time equation is (1) and the scalar equation gives $3(\dot{H} + 2H^2) = \kappa(\rho_\Lambda - 4\rho_C)$. The same action can independently define a four-dimensional metric and scalar. With the specified probe couplings its homogeneous and spectral predictions agree with those of the compact description. Interpreting b as a circle radius requires further physical information [3].

Frame dependence and stabilization

Set $g_E = (b/b_*) g_J$, $a_E = a\sqrt{b/b_*}$, $M_*^2 = 2\pi b_*/\kappa$ and $\varphi = \sqrt{3/2} M_* \log(b/b_*)$. Then

$$H_E = \frac{H + K/2}{\sqrt{b/b_*}}, \qquad V_E = 2\pi b_*^2 \left(\frac{\rho_\Lambda}{b} - \frac{C}{b^6} \right).$$

At the tuned event the Einstein-frame Hubble rate is $H_E = -H_*/2$, while the three-torus expands in the Jordan metric. A frame transformation also changes the matter rulers; an interpretation of observed redshift requires a matter action. The Einstein potential has one stationary radius $b_s^5 = 6C/\rho_\Lambda$, a maximum, and $b_s/b_* = 16^{1/5}$. The inverse-fifth-power source has no finite-radius potential minimum. The corrected source in §5 has one. In both cases a stationary scalar potential and a stationary spatial volume are different conditions.

An optical cross-check

Take parallel rays along z^1 with constant offsets in z^2, z^3, y. Normalize their affine parameter by $dt/d\lambda = 1/a$. Their transverse proper three-volume is $\mathcal{A} \propto a^2 b \propto a_E^2$. From $R_{00} + R_{\hat{1}\hat{1}} = \kappa(\rho + p_x) = 0$ one finds $2\dot{H} + \dot{K} = HK - K^2$. Hence

$$\Theta = \frac{d \log \mathcal{A}}{d\lambda} = \frac{2H + K}{a}, \qquad \frac{d\Theta}{d\lambda} = -\frac{2H^2 + K^2}{a^2}.$$

Although $\dot{V}_* = \ddot{V}_* = 0$, the bundle still contracts with $\Theta_* = -H_*/a_*$. Its unique earlier maximum is located in §9. This bundle spans the circle direction. A point-source distance requires different optical data [13,14].

An empirical exclusion with specified scope

Suppose present-day ordinary matter couples universally and minimally to g_J, the scalar is unscreened and effectively long-range on Solar-System scales, and the local background varies negligibly during measurement. In this weak-field limit the action has Brans--Dicke parameter $\omega = 0$ and predicts $\gamma_{\rm PPN} = (\omega + 1)/(\omega + 2) = 1/2$ [4]. Cassini radio tracking found $\gamma_{\rm PPN} = 1 + (2.1 \pm 2.3)\,10^{-5}$ [5]. These values exclude this present-day matter completion. The homogeneous equations supply no local suppression of scalar forces. Cassini tests this present-day completion; no observation in this manuscript dates the early-epoch solution.

7. Two-line inference and calibration drift

Let $A = a_o/a_e$ and $D = b_o/b_e$. The subscripts label reception and emission. Take comoving source and receiver and two co-emitted, externally propagating, massless zero-mode carriers in the geometrical-optics regime. Correct other velocity and gravitational shifts, and use the same units for local clocks. The spatial translation symmetries conserve the external comoving wave vector, so both frequencies scale by $1/A$ on the path.

Proposition 2. Calibration of relative volume

Suppose the reference frequency at each end is $m > 0$ and the circle-sensitive transition has local frequency g/b, with $g > 0$. Allow m and g to depend on the epoch. Write $\mu = m_o/m_e$, $\xi = g_o/g_e$, and $f_{\mathrm{ref}}, f_{\mathrm{sig}}$ for the two received frequencies. The volume estimator formed under the hypothesis of stable emission standards satisfies

$$\widehat{A} = \frac{m_o}{f_{\mathrm{ref}}} = A\mu, \quad \widehat{D} = \frac{f_{\mathrm{sig}}/f_{\mathrm{ref}}}{(g_o/b_o)/m_o} = D\mu/\xi, \qquad \widehat{A}^3\widehat{D} = \frac{V_o}{V_e}\frac{\mu^4}{\xi}. \tag{7}$$

Proof. The received frequencies are m_e/A and $g_e/(b_e A)$. Substitution yields the first two identities. The last uses external isotropy and $V_o/V_e = A^3 D$. The frequency drifts are relative to the common proper-time standard; changing that standard also requires a matter model. ∎

If independent calibration gives $|\log\mu| \le e_m$ and $|\log\xi| \le e_g$, the error in inferred log-volume due to these two drifts is bounded by $4e_m + e_g$. This calibration envelope is sharp if the two log drifts can attain opposite extremes together. It does not include counting noise, peculiar motion or uncertainty in the epoch. If an error also changes the inferred A, it moves the horizontal coordinate x used in the volume tests.

A single line identifies relative shape

Dividing the first two observables in the order $\widehat{D}/\widehat{A}$ gives

$$F := \frac{f_{\mathrm{sig}}}{g_o/b_o} = \frac{\widehat{D}}{\widehat{A}} = \frac{D}{A\xi}.$$

If g is stable, this single received-to-local frequency ratio identifies D/A, the change in circle size relative to one external length, for arbitrary reference drift μ. More generally $|\log\xi| \le e_g$ gives $D/A \in [Fe^{-e_g}, Fe^{e_g}]$. With ξ fixed, replacing (A, D, μ) by $(cA, cD, \mu/c)$ for any $c > 0$ preserves both line estimators and multiplies the geometric volume ratio by c^4. This freedom belongs to the optical observation map before imposing a specific cosmological history.

Numerical example

At $\tau_e = 0$ and $\tau_o = 0.05$, direct integration gives $A = 1.07215310$, $D = 0.80766835$, and $V_o/V_e = 0.99541340$. For stable standards choose emitted frequencies 1 and 1.5 in reference units. The received values are 0.93270 and 1.39905; the local values at reception are 1 and 1.8572. Their unrounded circle-sensitive ratio is $F \simeq 0.75331438$. With g fixed, a change of the reference frequency by a factor $(A^3 D)^{-1/4} \simeq 1.00114995$ yields an apparent volume ratio of one for the same geometric history. The example accommodates discrete external momenta. Choose harmonics $2N, 3N$ along one external direction, $m_e = 4\pi N/a_*$ and $g_e/b_* = 6\pi N/a_*$ with a positive integer N. This matches the ratio without specifying a transition or a linewidth. The receiver must lie on the signal path. No emitter, drift or actual spectra have been measured for this model. Astronomical comparisons of different spectral transitions similarly require sensitivity to their local frequency standards [11].

8. Limits of stabilization and emission

Passage through the inner minimum

For the exact corrected source with $0 < d < 5/6$, take the homogeneous flat Einstein metric and its single canonical scalar. Write t_E for its proper time. Equations in section 6 give $\varphi = \sqrt{3/2}M_* \log(b/b_*)$, $U = 2\pi b_*^2 \rho(b)/b$ (the Einstein-frame potential here), and

$$3M_*^2 H_E^2 = \tfrac{1}{2}\dot{\varphi}^2 + U, \qquad \dot{H}_E = -\dot{\varphi}^2/(2M_*^2), \qquad H_{E,*} = -H_*/2.$$

Dots in this paragraph denote t_E derivatives. Initially $\dot{\varphi}_* = -\sqrt{27/2}\, M_* H_*$ and $U_* = -6M_*^2 H_*^2$. From b_* down to the inner minimum, $U \le U_*$. Throughout this interval $H_E \le -H_*/2$ and $|\dot{\varphi}| \ge \sqrt{27/2}\, M_* H_*$. The initial inward sign persists. In field coordinates,

$$|dH_E/d\varphi| = \sqrt{(3M_*^2 H_E^2 - U)/(2M_*^4)}.$$

This grows at most linearly in $|H_E|$ on the compact field interval. Thus H_E stays finite there and the smooth solution continues through the minimum. Strict inequality in the interior gives

$$H_*(t_{E,\min} - t_{E,*}) < \tfrac{1}{3}\log(b_*/b_{\min}).$$

At the crossing the speed exceeds its initial magnitude. The flat Friedmann constraint excludes a vacuum at rest there, and monotonicity keeps H_E negative in this sector. The lifetime bounds in §9 require the original inverse-fifth law. See also Felder et al., Cosmology with negative potentials.

Kinematics of an emitted line

In a local adiabatic emission calculation, an isolated free, externally at-rest mode has energy $|n|/b_e$. For final free tower modes with conserved total compact momentum $\sum_j n_j = n$, energy satisfies

$$\sum_j E_j \;\ge\; \sum_j |n_j|/b_e \;\ge\; |n|/b_e.$$

A further $n = 0$ signal of positive energy is kinematically forbidden. Positive energy for a transition would have to come from additional source degrees of freedom. For an isolated two-body decay at external rest, with a massless zero-mode carrier, its energy is $(M_i^2 - M_f^2)/(2M_i)$ including recoil. A matter model must reproduce the required gap after this correction. Circle translations conserve the compact quantum number. Appelquist, Cheng and Dobrescu.

Other probe channels

For a periodic massless probe the local adiabatic dispersion is $E_{n,\mathbf{k}}^2 = (n/b)^2 + (2\pi|\mathbf{k}|/a)^2$. The first compact level at external momentum $\sqrt{3}/b$ and the second at rest share energy $2/b$ only if $|\mathbf{k}|^2 = 3[a/(2\pi b)]^2$. Resolving $E^2 - p^2$ can distinguish levels whose squared rest energies are in ratio $1 : 4$. Source bandwidth, allowed torus momenta and efficiencies enter any test. For a plane, source-free tensor zero mode, $d(a^3 b\,\dot{h})/dt + a^3 b\,\omega^2 h = 0$, $\omega \propto a^{-1}$, yields adiabatic strain $|h| \propto 1/(a\sqrt{b})$. A four-dimensional field model with matched masses and couplings can reproduce these listed responses. *Causal access.* Along an external coordinate, a null signal covers at most $\int_{t_p}^{t_s} dt/a < 0.61855/(H_* a_{\min})$. The external period is one in this coordinate, and a_* is free. Thus for $H_* a_{\min} \ge 0.61855$ a complete circuit is impossible even over the formal lifetime.

9. Maximal homogeneous history

Proposition 3. Conditional finite lifetime and turning order

Use five-dimensional comoving proper time t and let $\tau = H_*(t-t_*)$. For the source with $\rho_C = -Cb^{-5}$, set $\ell = -\log(b/b_*)$, $q = (H/H_*)^2 e^{-5\ell}$, $D = (16 - 6e^{-5\ell})/5$. On the future contracting-circle branch the constraint and the external equation yield

$$\frac{dq}{d\ell} = \frac{q}{q+D}\left[\frac{48+42e^{-5\ell}}{5} - 9q\right], \quad \frac{d\tau}{d\ell} = e^{-5\ell/2}\frac{\sqrt{q}}{q+D}, \quad \frac{dx}{d\ell} = \frac{q}{q+D}.$$

Here $K/H = -(q+D)/q$; the outward radius variable can be used without a turning point on this branch. Initially $q = 1$. The vector field points inward or is tangent at both ends of $1 \le q \le 2$. Thus the circle shrinks to zero in finite proper time. Bounding $d\tau/d\ell$ on this strip and integrating gives the future enclosure below. In reverse proper time, an outward-rounded integration to $\tau = -3/8$ certifies entry into the same strip with contracting circle in the reversed evolution. The same tail argument gives the past enclosure. For large ℓ the vector field points strictly toward $16/15$ outside any fixed neighborhood of this value, uniformly within the strip. At either end $q \to 16/15$, so $dx/d\ell \to 1/4$. Consequently $a \to \infty$ and $V \to 0$. The Ricci scalar is $10\kappa\rho_\Lambda/3$, while

$$R_{AB}R^{AB} = \kappa^2(20\rho_\Lambda^2/9 + 20\rho_C^2) \to \infty.$$

Event in forward time	Certified interval for τ
Past curvature singularity	$(-0.50, -0.45)$
Global volume maximum	$(-442/1024, -440/1024)$
Parallel-bundle transverse maximum	$(-90/256, -89/256)$
Circle radius maximum	$(-45/256, -44/256)$
Maximum of circle-to-external length ratio	$(-41/256, -40/256)$
External length minimum	$(-33/256, -32/256)$
Stationary volume inflection	0 by definition
Future curvature singularity	$(0.10627, 0.11855)$

At $H = 0$, (2) gives $\dot{H} > 0$. At a zero of K in the validated central interval $-3/8 \le \tau \le 0$ the bound $\kappa\rho_C/H_*^2 < -3/5$ gives $\dot{K} < 0$. The tail strip has $HK < 0$. Together with the isolated brackets in the table these signs give one external minimum, one circle maximum and two crossings of b_*. The identity $\ddot{V}/V = \kappa(8\rho_\Lambda/3 + \rho_C)$ gives strict convexity between the crossings and strict concavity outside them. Since $\dot{V}(t_*) = 0$ at the later crossing, $\dot{V} < 0$ throughout the intervening interval and thereafter. Before the first crossing the volume rises from zero and its derivative decreases strictly, giving one maximum. Its numerical estimate is $1.41426435\, V_*$. At the four-volume maximum $2H + K = -H > 0$, while at the circle maximum $2H + K = 2H < 0$. By monotonicity in §6 there is precisely one transverse maximum between them. It is at $\tau \simeq -0.349334886$; its certified bracket above follows from the signs of $2H + K$ at the two endpoints. A matching interval integration is embedded in this PDF. The full bounds are also in the global existence and singularity bounds, §§4--7. The hierarchy $|H|b \ll 1$ used to motivate the stress fails on both tails, since $|H|b = H_* b_* e^{3\ell/2}\sqrt{q}$.

Relative shape. Equation (2) gives $d[V(K-H)]/dt = 5\kappa V\rho_C < 0$. Since $d\log(b/a)/dt = K - H$, the ratio b/a has a unique maximum between the circle maximum ($K - H > 0$) and the external minimum ($K - H < 0$). Numerically $\tau \simeq -0.15719932$ there. At the volume inflection the ratio is already falling at fractional rate $-4H_*$. This is the shape probed by the single stable line in §7.

10. Information in five volume readings

Fix $r = 1/100$ and $h = r/2$. The record specifies the log-volume relative to its central value at $x_i = (i-2)h$, $i = 0, \dots, 4$. Interpret each x_i as an *external length*, not as a time. The pivot and length ratios are exact here. Inferences about the sign of the instantaneous slope depend on the allowed pressure laws.

Proposition 4. Curvature signs from the record

For any C^2 log-volume curve through the five tuned readings, there is a point in $(-r, 0)$ with $R'' > 0$ and a point in $(0, r)$ with $R'' < 0$. Both conclusions persist under arbitrary errors bounded by 3.97752 ppm in each log-volume, at the same abscissae.

Proof. Define $B_- = R_0 - 2R_1 + R_2$ and $B_+ = R_2 - 2R_3 + R_4$. For each triple on a uniform grid, $B = h^2 R''(\zeta)$ at some interior ζ by quadratic interpolation and two applications of Rolle's theorem. Integration of the certified series (3) gives

$$20.72268 < 10^6 B_- < 20.72269, \qquad -15.91009 < 10^6 B_+ < -15.91008.$$

The sum of the absolute coefficients of either difference is four, so perturbing every datum by at most ϵ changes either difference by at most 4ϵ. These enclosures imply the stated signs. ∎

x_i	-0.010	-0.005	0	0.005	0.010
$10^6 R_i$, rounded	27.08235	3.17983	0	−2.84094	−21.59197

In ordinary volume, the successive fractional losses $1 - \exp(R_{i+1} - R_i)$ are 23.90, 3.18, 2.84 and 18.75 parts per million. Each step increases the external length by the same factor e^h. The record forces a change in the pattern of loss per step. In ordinary ratios the same record has a geometric interpretation. In axes $a/a_*, V/V_*$ the four secants, multiplied by 10^6, lie in $(-4817, -4816)$, $(-638, -637)$, $(-567, -566)$ and $(-3723, -3722)$, respectively. These follow by exponentiating the same rational enclosures with Taylor remainders. The mean value theorem applied to successive secants again forces opposite curvature signs. Neither description fixes the maximum slope or its sign at the pivot. The allowance concerns the true log-volume vector relative to the tabulated values. For uncertain estimates, both signs must hold over a joint region for the true readings and abscissae. Appendix A gives an exact test when reference drift moves both coordinates.

Testing specified predictions

For the four outer readings, $K_4 = (R_0 - 2R_1 + 2R_3 - R_4)/6$ lies between 6.10546 and 6.10547 ppm. Its weights annihilate every quadratic. The sum of their absolute values is one. Subtracting K_4 times their signs gives a four-vector in the quadratic evaluation space. The contrast bounds the maximum-norm distance below by K_4 and this construction attains it. Disjoint bounded-error sets require errors below half that distance.

At the five nodes the $c_4 = 282$ history exceeds the inverse-fifth-law prediction by approximately 153.616286, 8.437725, 0, 6.730377 and 97.425013 ppb, respectively. Bounded-error boxes around the fixed prediction vectors are disjoint if and only if the largest absolute gap exceeds twice the common error radius. Less than 76.808 ppb in the first reading suffices for this pair. The normalized contrast with weights $(1, -4, 6, -4, 1)/16$ cancels constants, quadratics and odd powers. Its gap is 11.89805--11.89806 ppb; less than 5.949 ppb per reading suffices. Using a contrast and using individual predictions answer different questions. Neither bound identifies an exact stationary event in the larger class below.

11. Nonidentifiability from finite volume samples

Theorem 3. Finite length-indexed samples admit an everywhere-falling alternative

Let $I = [-r, r]$ be a compact window on the tuned expanding branch, with $R'(x) < 0$ for $x \neq 0$ and $R'(0) = 0$. These premises hold at $r = 0.01$ by the enclosure in §3. For every finite set of distinct external log-lengths in I, there is a local externally isotropic Einstein solution with analytic total density $\tilde{\rho}(b)$, $\tilde{p}_x = -\tilde{\rho}$ and $\tilde{p}_y = -\tilde{\rho} - b\tilde{\rho}_b$, which has the same volumes at all sampled lengths and strictly negative volume velocity everywhere on I.

Proof. Adjoin 0 to the nodes if needed and put

$$Q(x) = x \prod_{x_j \neq 0} (1 - x/x_j), \qquad R_\epsilon = R - \epsilon Q.$$

The polynomial vanishes at every node and $Q'(0) = 1$. In a neighborhood of zero $Q' > 0$. On the compact remainder of I, R' is bounded above by a strictly negative number and Q' is bounded. Thus all sufficiently small $\epsilon > 0$ give $R'_\epsilon < 0$ throughout I. The polynomial tends to zero after multiplication by ϵ in any fixed finite derivative norm on I.

To realize this curve dynamically, put $p = R'_\epsilon$, $b = b_* e^{R_\epsilon - 3x}$, $K = H(p - 3)$ and $\kappa\tilde{\rho} = 3H^2(p - 2)$. External vacuum pressure gives $2\dot{H} + \dot{K} = HK - K^2$. Equivalently solve

$$\frac{d \log H}{dx} = A := \frac{p' + (p - 4)(p - 3)}{1 - p}, \qquad H(0) = H_* > 0.$$

The denominator stays positive and $b_x = b(p - 3) < 0$. Consequently H and the density as a function of b are analytic and $dt = dx/H$ is a regular increasing clock. For $T = \kappa b\tilde{\rho}_b/(3H^2) = [2A(p-2)+p']/(p-3)$, substitution gives $A+p = 2(p-2)+T$ and $(p-3)A+p'+p(p-3) = 2(p-2)-2T$. These are the external and circle equations, respectively. The constraint holds by construction. Since $dV/dt = VHp$, the physical volume has strictly negative slope throughout the window. ∎

The constitutive law can change. The sample fixes no elapsed times or derivatives. A direct local identification would require, for example, $\dot{V} = \ddot{V} = 0$ and $\dddot{V} < 0$ in comoving time. An isolated exact slope of zero alone is also compatible with an ordinary extremum. A fixed microscopic stress or independent timing defines another inverse problem. No assertion about arbitrarily long histories or arbitrary observed tables follows.

Explicit five-point example

For $\zeta = x/r$ use $R_\delta = R - \delta\zeta(\zeta^2 - 1/4)(\zeta^2 - 1)$ with $\delta = 30 \times 10^{-6}$. All five node values agree exactly. Its central slope is -0.00075. An outward rational enclosure over 256 subintervals gives $R'_\delta < -0.0004$ on all of I. The theorem supplies a dynamical realization. Even if further exact length-indexed readings are requested, there is a possibly different admissible alternative for each finite set. This finite result does not cover exact observations throughout an interval.

12. Physical interpretation and comparison

The present model specifies a homogeneous geometry, a source and tuned data. Its quantitative predictions concern that solution. No matter sector, thermal history, primordial perturbation spectrum, noise model or likelihood for cosmological data is supplied.

Published comparison models

Comparison	Distinct physical or mathematical commitment
Five-dimensional vacuum [1]	With $H > 0$, (1) at $\rho = 0$ gives $K = -H$ and $V/V_* = (a/a_*)^2$. The circle contracts while the total volume grows. Directional contraction alone fails to predict the event.
Higher-dimensional steady state [2]	Total volume and effective density are held constant as model conditions. In §4 an interval of exact compensation instead requires $\rho \propto -b^{-8}$ within our specified pressure class.
Casimir stabilization [9]	Greene and Levin exhibit field spectra whose Casimir contributions can stabilize toroidal dimensions. Here the particular massless circle term and positive constant produce a contracting circle past the event; the static finite-torus source itself needs correction (§5).
Base ΛCDM [10]	Planck fits temperature, polarization and lensing data with a six-parameter model and a perturbation spectrum. Its observables do not include this four-volume. An empirical comparison would need predictions for common calibrated data.

Dependence on the fields retained

In a reduced two-term model with positive density $A = \Lambda_s e^{10\phi/3}$, negative $-C/b^5$, and kinetic density $2\dot{\phi}^2/(3\kappa)$, the field can be at rest at the event. Its equation $\ddot{\phi} + \theta\dot{\phi} = -5\kappa A/2$ then gives $\ddot{\phi}_* = -9H_*^2$. Such a solution reproduces the cubic coefficient -24 and gives $c_4 = 258$, compared with 270 here and 282 for the shear example in §4. A two-epoch antisymmetric difference cancels even powers, so it cannot isolate this quartic difference. These conclusions are local to the specified field equations. A microscopic string compactification realizing the source and fixed spectator fields has yet to be derived [12].

Domain of the physical claim

A field state and its renormalized stress on the evolving finite torus are needed to justify the constitutive law. At the event $(H_* b_*)^2 = 5\kappa C/(48 b_*^3)$. The model allows both $a_* \gg 2\pi b_*$ and $H_* b_* \ll 1$; neither hierarchy fixes the stress. A physical interpretation of (7) also requires calibrated matter and an external signal. The four-dimensional reduced scalar with matched couplings reproduces the listed records. A microscopic source, a cosmic date and the identification of an extra direction are each open. Cassini already excludes the specified unscreened present-day matter completion (§6).

Appendix A

Calibration of the five volume readings

How accurately must two emission frequencies be known to infer the curvature signs in §10? The limits below apply to the specified five readings, exact line ratios and a known central anchor. All error bounds are deterministic.

A.1 Frequencies, readings and units

Compare each emission to the same reception epoch under the propagation law of §7. Normalize to the central emission, indexed by $i = 2$. For the ordinary reference frequency m and the coefficient g of the circle-sensitive line g/b, write

$$d_i = \log(m_i/m_*), \qquad e_i = \log(g_i/g_*), \qquad d_2 = e_2 = 0.$$

With hats denoting the readings inferred using the central standards, (7) gives

$$\hat{x}_i = x_i + d_i, \qquad \hat{R}_i = R_i + 4d_i - e_i. \tag{A1}$$

Take $\hat{x}_i = (i-2)h$, $h = 0.005$, and take $\hat{R}_i$ from the tuned curve of §10 before rounding. Each noncentral error can independently take any value within $|d_i| \le \delta$, $|e_i| \le \eta$. No distribution is assigned. One ppm means 10^{-6} in log-frequency.

A.2 Limiting errors

The left triple tests for increasing slope somewhere between its first and last true lengths. The right triple tests for decreasing slope on its corresponding interval. For $\delta < h/2$ the nodes stay ordered.

	Critical drift (ppm)	
Fixed condition and quantity varied	Left triple	Right triple
Circle-sensitive coefficient fixed ($\eta = 0$), vary δ	1.72602	1.32530
Ordinary reference fixed ($\delta = 0$), vary η	6.90756	5.30336
Reference drift bounded by 1 ppm, vary η	2.90554	1.30173

Values are rounded to the nearest displayed digit. Each sign holds for every allowed calibration strictly below its exact critical value. At equality an allowed calibration makes that triple collinear. For example, if both caps are 1 ppm, both signs follow. With a 1 ppm reference cap and a 1.4 ppm cap on g, the right-hand sign is no longer guaranteed. The exact boundary is given on the next page.

The 3.97752 ppm allowance in §10 is an error in *log-volume at known lengths*, allowing perturbations of all five ordinates. Here the central reading has zero error, the other coordinates can move, and the caps refer to the two *frequency standards*. No calibration or measured curve for this apparatus has been supplied.

Appendix A

Proof of the calibration bounds

A.3 Adjacent slopes

Put $S_i = \hat{R}_i - 4\hat{x}_i$. By (A1), $R_i - 4x_i = S_i + e_i$. Subtraction of $4x$ preserves differences of successive slopes. For a triple beginning at $j = 0$ or $j = 2$, define

$$u = S_{j+1} - S_j, \quad v = S_{j+2} - S_{j+1}, \quad s = h + d_j - d_{j+1}, \quad t = h + d_{j+1} - d_{j+2}.$$

The true adjacent widths s, t are positive. Multiplying the second secant slope minus the first by st gives

$$L = \underbrace{sv - tu}_{L_0(d)} + te_j - (s+t)e_{j+1} + se_{j+2}. \tag{A2}$$

By the second divided-difference theorem, the sign of L gives the sign of the second derivative at some interior point. The first derivative at the central reading is still undetermined.

A.4 Exact worst case

Let $\sigma = +1$ for the left triple and $\sigma = -1$ for the right. At fixed d, let $M(d)$ sum the absolute coefficients of the free e_i in (A2), omitting the anchor. The smallest signed numerator is $\sigma L_0(d) - \eta M(d)$. It is attained by taking each free e_i at its opposing endpoint. Since the widths are positive, $M(d)$ is affine in d. The remaining minimization over the two free reference drifts thus needs only their four corner assignments $d_i = \pm\delta$.

If reference drift alone preserves a strict sign, the maximum allowable η at the boundary is

$$\eta_j(\delta) = \min_{d \text{ at corners}} \frac{\sigma L_0(d)}{M(d)}. \tag{A3}$$

The minimum is attained and $M(d) > 0$. Both signs are guaranteed below the smaller critical value. If reference drift alone permits zero, allowing further error cannot restore a strict uniform sign.

For $\eta = 0$, expand $L_0 = h(v-u) + vd_j - (u+v)d_{j+1} + ud_{j+2}$. Let N_j sum the absolute coefficients of the free d_i. The exact reference threshold is $h|v-u|/N_j$. Outward rational enclosures in ppm are $(1.72601950, 1.72601951)$ for the left triple and $(1.32530103, 1.32530104)$ for the right. For $\delta = 0$, the two free coefficients of the circle-sensitive drift have total magnitude $3h$. This gives the second row of the table. The third row uses (A3).

The embedded calculation encloses the tuned data with rational Taylor residuals and evaluates (A2) at all 16 joint corners. These limits concern the two tests on this record. Inferring a stationary event or identifying its matter source requires information beyond the signs of curvature.

References and reproducibility

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Computational reproducibility

Four Python files are embedded in this PDF. Extract them to one directory and run them with Python 3. The local enclosures use the standard library. The numerical and symbolic checks also use NumPy, SciPy and SymPy.

`verify_stationary_compactification.py` certifies the local degree-24 expansions with rational residual bounds. `verify_reading_example.py` encloses the explicit everywhere-negative slope in §11.

`verify_volume_observations.py` gives the outward second-difference enclosures in §10, checks the line identities and Appendix A enclosures, and integrates the optical epoch in §7. `verify_global_dynamics.py` certifies the tail and turning-point brackets including the transverse maximum; it also checks the shear, finite-window and corrected potential results.

The §10 thresholds assume exact epoch coordinates; Appendix A treats correlated drift. The dimensional scale, frequency standards and detector uncertainty must be supplied independently. Numerical estimates are marked separately from rigorous enclosures. The public figures are calculated from (2) and the tail system in §9. For the explicit interpolant in §11, see the construction for five volume readings. The illustrated account gives the calculation in ordinary lengths and light frequencies.